SELF-MONITORING AND ANONYMOUS COUNSELLING BASED DIGITAL COMPANION FOR ACADEMIC STRESS MANAGEMENT

25-26J-480

Project Proposal Report

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Sri Lanka Institute of Information Technology Sri Lanka

August 2025

PERSONALIZED QUESTIONS ASSESS EMOTIONS, GIVING REAL-TIME COPING STRATEGIES FOR STUDENT STRESS

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate dissertation under my supervision.

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Signature of the supervisor   Date

(Prof. Samantha Thelijjagoda)

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ABSTRACT

Student stress and anxiety have become critical challenges in higher education, affecting academic performance, mental health, and overall well-being. While digital mental health (mHealth) applications such as Calm and Headspace provide coping strategies, they largely adopt a reactive approach, requiring students to self-identify stress before seeking support. This limitation creates a significant gap in proactive stress management and personalized intervention.

This proposal introduces a mobile-based solution that leverages personalized daily questions, AI-driven emotion detection, and predictive modeling to identify stress patterns in students and deliver timely coping strategies. Using the Experience Sampling Method (ESM), the system captures real-time emotional data through short, context-aware check-ins. Natural Language Processing (NLP) models analyze responses to detect subtle emotional cues, while Long Short-Term Memory (LSTM) networks forecast potential stress spikes. To provide effective interventions, the platform incorporates Just-in-Time Adaptive Interventions (JITAIs) and evidence-based Cognitive Behavioral Therapy (CBT) techniques, ensuring students receive proactive and personalized support.

An additional novel feature of the system is the integration of adaptive soundscapes, which generate real-time audio environments to aid concentration and relaxation based on the user's emotional state. The application is designed using React Native for cross-platform usability and supported by Firebase for secure, scalable data management.

The anticipated results include improved stress detection accuracy, higher student engagement through personalization, and measurable reductions in self-reported stress levels. This project aims to transform student mental health support from a reactive, one-size-fits-all model into a proactive digital companion, fostering resilience, well-being, and academic success.

Keywords - Academic Stress, Mental Health Technology, Artificial Intelligence (AI), Proactive Support, Experience Sampling Method (ESM), Recommender Systems, Causal Inference, Adaptive Soundscapes, Mobile Health (mHealth)

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
JITAI	Just-in-time adaptive interventions
CBT	Cognitive Behavioral Therapy
NLTK	Natural Language Toolkit
LSTM	Long Short-Term Memory
ESM	Experience Sampling Method

1. INTRODUCTION

For young adults, the move into higher education is a significant and often challenging transition. During this time, students face a mix of tough work, new social situations, and big personal changes. They have to deal with heavy workloads, strict deadlines, and constant pressure to get good grades, which creates an environment where stress can easily build. On top of their studies, many are also learning how to make new friends, manage money, and live independently for the first time. All of these factors together can lead to increased stress and anxiety, which can harm not only a student's academic success but also their general quality of life and future well-being [1].

To address this widespread issue, digital mental health apps have become a key source of support, valued for their wide reach and ease of use. The fundamental problem with most of these apps, however, is that they are designed to be reactive. They offer libraries of coping mechanisms and resources, but this design relies on students to first identify their own struggles and then actively look for help. This model places the entire burden on the individual, who may not have the presence of mind or the energy to reach out during a crisis. Consequently, a major gap in preventative care remains, allowing the quiet, day-by-day buildup of academic stress to continue unchecked until it becomes unmanageable.

My part of the project directly addresses this gap by shifting the focus from reactive to proactive emotional support. The system uses a smart AI engine to maintain a continuous, supportive dialogue with the user through personalized daily questions. Unlike a simple mood tracker, this tool is designed to be conversational and inquisitive. It analyzes the user's text responses to find subtle emotional patterns and predict rising stress levels. This foresight makes it possible to offer "just-in-time adaptive interventions" (JITAIs), which are targeted, evidence-based techniques like cognitive reframing or guided mindfulness, delivered at the exact moment they are most needed and likely to be effective [2].

A central innovation of this project is its "Adaptive Soundscapes." Going beyond fixed playlists of music or nature sounds, these are responsive audio environments that change in real-time according to the user's emotional state. The result is a personalized sound experience that can either help with focus or promote relaxation. By providing support that is both immediate and highly personal, the system does more than simply manage stress. It seeks to empower students by helping them build lasting emotional strength and develop healthy coping skills, which are vital for their long-term success in school and for their overall mental well-being [3].

2. BACKGROUND AND LITERATURE SURVEY

Background

The transition to university life represents a critical period in young adulthood, often accompanied by academic, social, and personal pressures. Studies indicate that this stage exposes students to heightened stress, anxiety, and emotional instability, directly impacting academic performance, concentration, and overall well-being. According to the WHO World Mental Health Surveys, nearly 31% of college students worldwide report symptoms of a mental disorder, with stress and anxiety being the most common [1]. The American College Health Association (2021) further reported that over 60% of students experienced overwhelming anxiety, while 45% reported feeling so depressed it was difficult to function. These statistics underscore the urgent need for effective, scalable mental health interventions tailored to the academic context.

In parallel, digital technologies, particularly mobile health (mHealth) applications, have emerged as promising tools for addressing these challenges. Apps such as Calm, Headspace, and BetterHelp offer guided meditation, stress reduction exercises, and counselling services. Studies demonstrate their efficacy in reducing stress and enhancing mindfulness among university students [2], [3]. Their key strength lies in accessibility, affordability, and scalability compared to traditional counselling services, which are often limited by resource constraints.

However, despite their popularity, existing apps suffer from structural limitations. They rely on a reactive model: students must first identify stress and consciously engage with the app. Many students fail to do so, particularly during periods of acute academic pressure, leading to missed opportunities for early intervention [4]. In addition, coping strategies are typically generic, assuming that all users benefit equally from the same set of practices. In reality, stress triggers vary exams, deadlines, public speaking, or social isolation yet interventions are rarely personalized [5]. This reduces long-term user engagement, as students perceive the content as repetitive or irrelevant to their lived experiences.

Furthermore, research shows that disengagement rates are high. A study by Mohr et al. [6] found that a significant percentage of users discontinue app usage within weeks, highlighting issues of sustainability and adherence. As a result, while current solutions show promise, they fall short in delivering continuous, individualized, and proactive support.

To address these gaps, there is growing consensus that digital interventions must evolve into intelligent, adaptive systems. Instead of waiting for stress to escalate, they should leverage real-time data collection, predictive modelling, and proactive coping mechanisms to anticipate challenges and intervene earlier. Emerging techniques such as personalized daily questioning, emotion recognition, and adaptive soundscapes present opportunities to build systems that not only manage but also prevent stress escalation, supporting both academic achievement and long-term resilience.

This proposal builds on these insights by designing a proactive AI-powered digital companion that integrates personalized daily questions, predictive models (LSTM + NLP), and adaptive soundscapes to provide timely, context-aware interventions. By shifting the paradigm from reactive to proactive, the system seeks to empower students to take control of their mental health while maintaining focus, productivity, and well-being.

Literature Survey

The body of literature on digital mental health interventions reveals both encouraging evidence and critical gaps that motivate the proposed solution.

1. Existing mHealth Solutions

Commercial apps such as Calm, Headspace, and IntelliCare have shown positive outcomes in stress management. Huberty et al. [2] demonstrated that the Calm app reduced stress among students in a randomized controlled trial, while Lattie et al. [3] confirmed benefits of mobile mental health interventions for anxiety and depression. However, these tools primarily act as passive repositories of content, requiring users to initiate usage when already stressed. This reactive paradigm diminishes their preventative potential.

2. Shortcomings of Current Approaches

Literature consistently identifies several shortcomings. First, interventions are reactive, offering support only after stress escalates [4]. Second, apps rely on generalized content guided meditations, relaxation tracks, or journaling prompts rather than personalized support. This mismatch often leads to weak engagement and abandonment [6]. Finally, most systems lack the ability to forecast stress patterns, leaving students vulnerable during high-pressure events like exams or project deadlines [7].

3. Personalization & Digital Therapeutic Alliance

Studies emphasize that personalization enhances user trust and adherence. Kretzschmar et al. [8] found that users perceived conversational agents as more effective when they could adapt to prior interactions, forming a therapeutic alliance akin to a relationship with a counsellor. This supports the argument that future digital tools must not only deliver content but also foster rapport and continuity with users.

4. Real-Time Data Collection (ESM/EMA)

Traditional surveys rely on retrospective self-reporting, which is prone to memory bias. In contrast, the Experience Sampling Method (ESM) and Ecological Momentary Assessment (EMA) gather emotional data in real time, producing richer, more accurate insights [9]. For student populations, daily personalized questioning is a practical and low-burden adaptation of ESM, enabling continuous tracking of emotional states and stress levels.

5. Predictive Modeling & Adaptive Interventions

Machine learning models, particularly Long Short-Term Memory (LSTM) networks, excel at analyzing sequential data, making them ideal for forecasting stress escalation. Wang et al. [7] demonstrated the feasibility of using smartphone data and ML models to assess student mental health trends. Coupled with Natural Language Processing (NLP), which analyzes emotional tone in daily responses, these methods allow systems to move beyond monitoring into proactive forecasting.

Building on predictions, Just-in-Time Adaptive Interventions (JITAIs) can deliver context-aware coping strategies such as guided breathing or mindfulness at the right moment. Interventions grounded in Cognitive Behavioral Therapy (CBT) have shown robust evidence in reducing anxiety and building resilience [10].

6. Novel Contributions in Literature

A particularly innovative trend is the use of adaptive soundscapes, where audio environments dynamically adjust to emotional states. Research in psychoacoustics demonstrates that music and sound can significantly regulate mood, concentration, and relaxation [11]. Yet few mental health apps integrate this feature in real time. By combining adaptive audio with predictive AI, the proposed system addresses an underexplored opportunity in digital well-being.

3. RESEARCH GAP

While there are several well-known digital mental health apps available, a closer look shows they have major gaps in providing students with complete, proactive, and truly personal support. By examining popular systems like Calm, StudentLife, IntelliCare, and Sensa, we can pinpoint the specific limitations our proposed solution is designed to overcome. Each of these apps offers valuable tools, but none provide the integrated, forward-looking approach needed to effectively prevent and manage academic stress. The following section explores the specific gaps left by these current tools.

Table 1 Comparison of the existing research methods and proposed method

Feature	[2]	[7]	[4]	[6]	Our Solution
Proactive Emotional Pattern Recognition	✕	✔	✕	✕	✔
Personalized Daily Questioning Mechanism	✕	✕	✕	✔	✔
Real-time Adaptive Coping Strategies	✕	✕	✔	✔	✔
Adaptive Soundscapes	✕	✕	✕	✕	✔
Proactive Notifications	✕	✕	✕	✕	✔

Calm App

The Calm app is a good example of a useful tool. A study by Huberty et al. (2019) found that its large library of mindfulness and meditation content helps reduce stress for college students [2]. The app is excellent at offering on-demand coping strategies that users can access when they feel overwhelmed.

However, Calm's biggest limitation is that it's reactive. It puts the responsibility on users to recognize they need help and then manually search for a suitable session. The system doesn't learn from a user's history to predict future stress or send preventative warnings. It also doesn't use personalized daily questions to gather emotional data, and it doesn't include the adaptive soundscapes. Our solution is designed to fill these specific gaps by building a proactive system that anticipates what a user needs and offers support, like adaptive soundscapes, before they have to look for it.

StudentLife

The StudentLife project by Wang et al. (2014) was a major step toward proactive mental health assessment [7]. By passively collecting data from phone sensors, it successfully identified behavioural trends and connected them to mental health and grades. Its key strength is its ability to recognize emotional patterns without needing the user to actively do anything.

However, the system is mainly a tool for assessment, not for providing direct help. It lacks a solid way to offer real-time, adaptive coping strategies based on its findings. Furthermore, it doesn't engage users with daily questions that could provide important emotional context that sensor data alone can't capture. Without interactive tools like adaptive soundscapes or proactive notifications for self-management, it remains mostly an analytical tool. Our project builds on the proactive idea of StudentLife but completes the circle by connecting these predictive insights with immediate, actionable, and personalized support.

IntelliCare

The IntelliCare suite, as described by Mohr et al., takes a skills-based approach to helping with depression and anxiety by using a collection of different "mini-apps" [4]. Its main strength is that each small app is designed to teach a specific coping skill, letting users practice adaptive strategies in real time. This modular design is good for skill-building.

However, much like Calm, IntelliCare is reactive; it doesn't analyze emotional patterns over time to predict when a user might need help or send alerts. The system is also missing a unified daily check-in to tie the experience together and collect emotional data long-term. Additionally, it doesn't use new engagement tools like adaptive soundscapes. Our proposed system will bring these kinds of adaptive coping strategies together into a single, intelligent platform that also has the proactive and personalized features that IntelliCare lacks.

Sensa Mobile App

The Sensa mobile app, studied by Valinskas et al. (2023), is a more recent tool that uses personalized daily questions to help manage stress and anxiety symptoms [6]. It also provides coping strategies in real time, making it a strong modern competitor.

However, its main drawback is that it lacks a truly proactive analysis system. Although it collects daily data, it doesn't appear to use this information to predict patterns and warn users about potential downturns in their mood. The app is also missing proactive notifications and innovative, dynamic features like adaptive soundscapes, using more traditional content libraries instead. Our solution improves on the idea of daily questioning by using this data to power a predictive model. This creates a truly proactive support system that is also enhanced with new, engaging, and highly personalized tools.

4. RESEARCH PROBLEM

The mental health and well-being of university students has become an increasingly urgent issue, with stress and anxiety emerging as the most significant concerns. The World Health Organization (WHO) reports that nearly one-third of university students globally experience symptoms of a diagnosable mental disorder, often related to stress and academic pressure [1]. These psychological challenges not only hinder academic performance but also contribute to absenteeism, reduced concentration, and in severe cases, withdrawal from studies. While traditional counselling services provide valuable support, their reach is limited by resource constraints, stigma, and accessibility issues. As a result, many students remain underserved, creating a pressing need for scalable, technology-driven solutions.

Digital mental health interventions, particularly mobile health (mHealth) applications, have grown rapidly in recent years. Apps such as Calm, Headspace, and IntelliCare offer meditation, mindfulness, and stress-reduction exercises [4]. Evidence demonstrates that these tools can provide short-term relief and improve user well-being [2]. However, their current design models introduce substantial limitations. Firstly, they operate on a reactive basis, requiring students to self-identify stress and initiate app usage [3]. Research shows that students often fail to engage at the right time due to lack of awareness, motivation, or overwhelming workloads [6]. Consequently, these tools are unable to prevent stress escalation and tend to deliver support too late.

Secondly, most existing solutions adopt generic coping strategies that overlook individual differences in stress triggers. For example, a student struggling with exam anxiety may require different interventions compared to one facing social isolation or time-management issues [8].

Thirdly, there is limited integration of predictive modeling and adaptive interventions. Few applications leverage advanced Artificial Intelligence (AI) methods, such as Long Short-Term Memory (LSTM) networks for temporal stress prediction or Natural Language Processing (NLP) for emotion detection in student responses [7]. As a result, opportunities to provide proactive, forecast-driven interventions are underutilized.

Finally, novel modalities such as adaptive soundscapes which dynamically adjust audio environments to support focus and relaxation are underexplored in the literature. Although psychoacoustic research demonstrates the effectiveness of sound in mood regulation [11], mainstream mHealth platforms have yet to integrate this feature into real-time student support systems.

In light of these limitations, the core research problem can be articulated as follows:

How can a proactive, personalized, AI-driven digital companion be developed to detect, predict, and manage student stress in real time, thereby overcoming the reactive, generic, and non-predictive nature of existing mHealth solutions?

Addressing this problem involves designing a system that integrates personalized daily questioning, predictive AI models (LSTM + NLP), just-in-time adaptive interventions (JITAIs), and adaptive soundscapes to create a comprehensive platform for proactive stress management. By tackling the shortcomings of current tools, the proposed solution aims to reduce stress levels, increase engagement, and promote long-term student resilience.

5. OBJECTIVES

5.1 Main Objective

The main goal of this research is to design and build an advanced, AI-powered emotional support system. It is specifically made to change the approach of digital mental health from reactive to proactive. The core idea is to create a digital companion that actively supports the mental well-being of young people by moving beyond simple tracking to offer smart, preventative help. To do this, we will develop a personalized daily questioning feature that will serve as the main data-gathering tool. This will allow the system to build a deep, contextual understanding of each user's individual emotional patterns.

This basic understanding is what will drive the system's main feature: providing support that is personal and relevant to the user's situation in real time. The goal is to create a smooth, continuous loop where the system analyses subtle changes in a user's feelings and instantly provides real, helpful support. This support will be delivered in two main ways: through proven coping skills that are offered at just the right moment, and through soundscapes that change on their own to help the user focus or relax. In the end, our aim is to build a complete system that does more than just help students through moments of high stress. It will also give them the tools and understanding they need to build lasting emotional strength.

5.2 Specific Objectives

1. **To develop an emotional data prediction engine models.**

This objective establishes the foundational data layer of the system, encompassing both the collection and analysis of longitudinal emotional data. This involves extensive research into validated short-form psychological scales and questioning techniques suitable for momentary assessment of stress, mood, and context in student populations. The questions will be designed to be quick to answer, and capable of capturing the nuanced fluctuations in a student's emotional state throughout the academic cycle. Given the sequential and time-dependent nature of mood data, this task is well-suited for advanced machine learning model. The implementation of this model will fulfill the task of developing a system that uses this analysis to "trigger proactive alerts or suggestions," directly realizing the novelty of "Proactive Emotional Pattern Recognition".

2. **To develop a recommendation engine for delivering personalized, evidence-based coping strategies in real-time.**

This objective addresses the core intelligence of the system: its ability to recommend effective interventions. An intervention system must go beyond correlation to estimate the causal effect of its recommendations. The engine will map the user's predicted emotional state (from Objective 1) to a curated library of evidence-based coping strategies, such as techniques from Cognitive Behavioral Therapy (CBT). The system can provide "Real-time, Adaptive Coping Strategies" that are not just personalized but are optimized for interventional impact.

3. **To develop a adaptive soundscape for focus and relaxation by with real-time emotional state data.**

The objective is use of static, pre-recorded audio libraries for relaxation and instead proposes a dynamic, generative approach. The work will begin with research into the field of psychoacoustics to understand the effects of different auditory characteristics such as frequency, tempo, timbre, and complexity on human psychological states, including stress, anxiety, and concentration.

4. **To develop, build, and evaluate a user-centric mobile interface that ensures high engagement, and ethical data handling.**

The design of the mobile interface is critical to the success of the entire system, as it mediates every interaction and data point. The design process will adhere to established UI/UX best practices for mental health and wellness applications, prioritizing a calming visual aesthetic, intuitive navigation, and minimal cognitive load. The intervention's effectiveness in mitigating academic stress will be evaluated by administering validated psychological scales such as the Depression, Anxiety and Stress Scale, and the WHO-5 Well-Being Index—at baseline and post-intervention to measure changes in key mental health indicators.

6. METHODOLOGY

The project's methodology is based on a strong, tiered architecture that maintains the mobile application that the user actually works with isolated from the complex AI processing that happens in the back-end. Such an arrangement guarantees the user has a responsive and seamless experience, but, simultaneously, allows for strong data analysis and support generation. The entire system runs on a sustained feedback loop: user input guides the AI, and the AI's output is then delivered back to the user as customized support.

6.1 System Architecture and Technologies

The overall architecture consists of a mobile client, a secure backend, and a suite of AI microservices.

- **Mobile Application (Frontend):** The user interface will be a mobile app created with React Native. This technology allows the app to work on both iOS and Android phones from a single codebase, which makes the development process more efficient and ensures a consistent experience for all users. The app itself will be responsible for showing the daily questions, capturing text responses, displaying coping strategies, and playing the adaptive soundscapes. A secure REST API will handle all communication between the app and the backend system.
- **Backend and Database:** The backend system will be built on Firebase, a platform from Google. Firebase provides a set of tools that are a great fit for this project. We will use Firestore as the main database to store user data, emotional history, and application content. Firebase Authentication will be used to handle secure user sign-ups and logins, while Firebase Cloud Functions will provide the serverless environment to run the AI Decision Engine and manage the API. This serverless approach is very scalable and cost-effective, as it automatically handles computing resources based on demand.
- **AI Engines:** The core AI functionalities will be developed as separate microservices using Python, which is the industry standard for machine learning and data science due to its extensive libraries and frameworks. This microservice architecture allows each AI component to be developed, tested, and scaled independently. Key Python libraries that will be utilized include TensorFlow and PyTorch for building and training the deep learning models, Scikit-learn for classical machine learning algorithms, and NLTK (Natural Language Toolkit) for text preprocessing. These services will be containerized using Docker and deployed on a cloud platform, communicating with the Firebase backend.

6.2 Data Flow and Process

The system's operational process begins with the user and flows through the various architectural components:

- **User Interaction**: The process begins when a user opens the mobile app and receives a personalized daily question. After they type their answer, it is sent securely to the Firebase backend.
- **Backend Processing**: Upon receiving the text input, a Firebase Cloud Function is triggered. This function saves the raw text to the user's profile in Firestore and then makes calls to the relevant Python-based AI microservices for analysis.
- **AI Analysis**:
  - **Real-time Emotion Analysis Model**: The user's text is then sent to an emotion analysis service. This service will use a pre-trained model, like a version of BERT, to analyse the sentiment and figure out the user's immediate emotional state (such as happy, sad, anxious, or stressed). The model is designed to understand the context and subtleties of language, which makes it more accurate than simpler methods that just look for keywords. To protect user privacy, we will first look into whether we can use tools like TensorFlow.js to run a simpler version of this analysis directly on the user's phone.
  - **Proactive Pattern Recognition Engine**: This service then pulls the user's past emotional data from Firestore. It will use a Long Short-Term Memory (LSTM) neural network, a tool specifically designed to handle data in a sequence. LSTMs are very effective at spotting long-term patterns and connections over time. The model will analyse the sequence of daily emotional states to predict the likelihood that the user's stress will significantly increase or their mood will decline in the near future.

6.3 Application of Knowledge Domains

- Artificial Intelligence (AI) & Machine Learning (ML): Emotion prediction, pattern recognition.
- Natural Language Processing (NLP): Analysing student responses for emotional context.
- Mobile & Cloud Computing: Uses React Native for cross-platform app development and Firebase for storage, authentication, and real-time updates.

6.4 Expected Outcomes

- AI models (NLP for emotion detection, LSTM for stress forecasting) capable of achieving $\geq$ 95% accuracy, precision, and recall using benchmark datasets (RAVDESS, Emo-DB, SER) and real-world student inputs.
- Integration of Just-in-Time Adaptive Interventions (JITAIs) and Cognitive Behavioral Therapy (CBT) techniques, ensuring students receive tailored coping mechanisms (e.g., guided breathing, reframing exercises).
- Pilot testing expected to show reduced perceived stress and increased concentration (validated by DASS and WHO-5 psychological scales).
- A cloud-enabled backend (Firebase) ensuring data security, scalability, and real-time performance, compliant with ethical clearance requirements.

6.5 System Architecture

System Overview

The system begins with the user interacting through the mobile interface, where they provide inputs such as daily Q&A, quantitative ratings, and calendar events. These inputs are sent via the REST API and processed in the data preparation stage before being stored in the database. The prepared data is analysed through three main engines: the real-time emotion analysis model, proactive pattern recognition, and the soundscape engine. Based on this analysis, the system delivers outputs including real-time coping strategies, adaptive soundscapes, personalized insights and visualizations, and proactive notifications, all of which are fed back to the user through the interface.

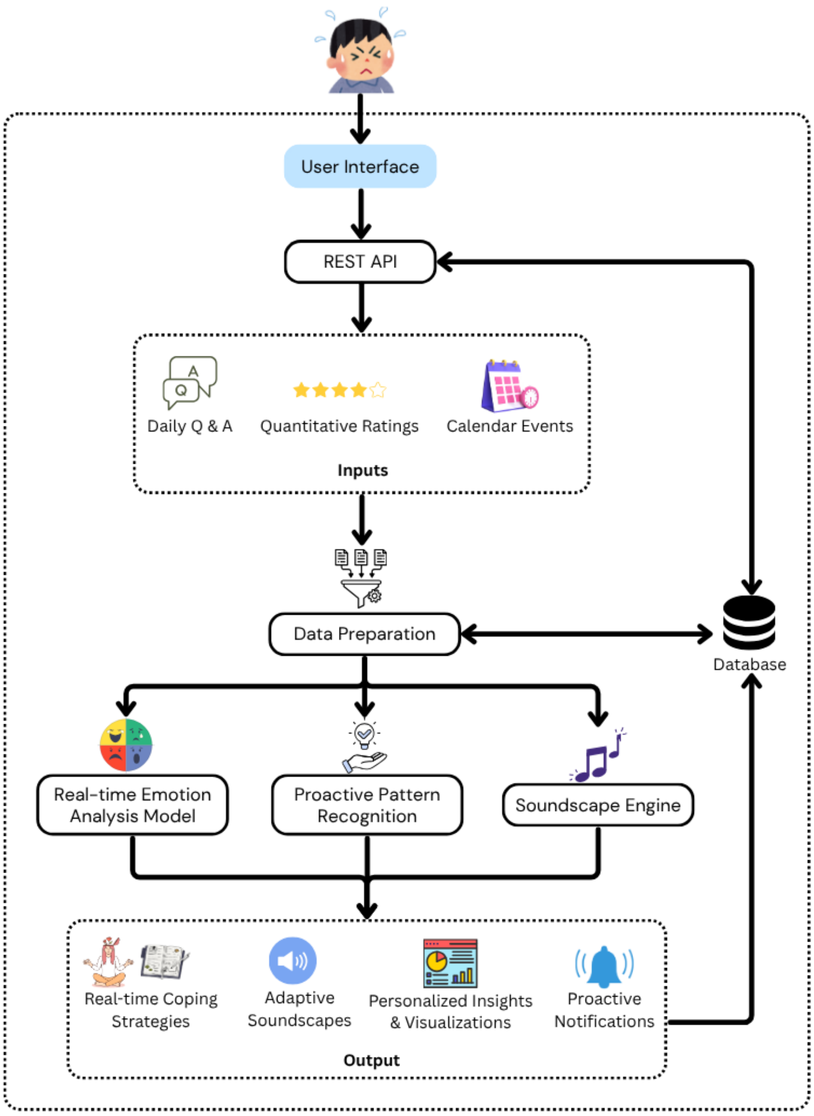

Figure 1 System Architecture

7. GANTT CHART

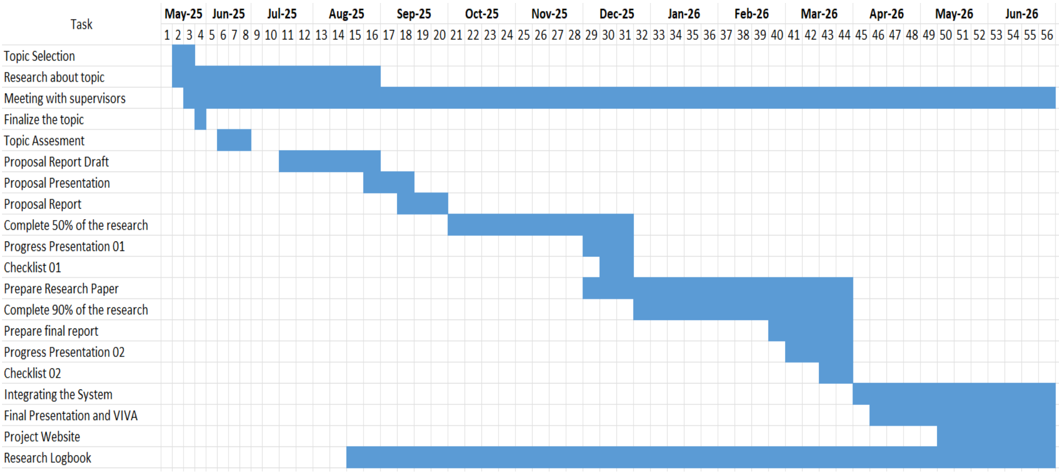

Figure 2 Gantt Chart

8. WORK BREAKDOWN STRUCTURE(WBS)

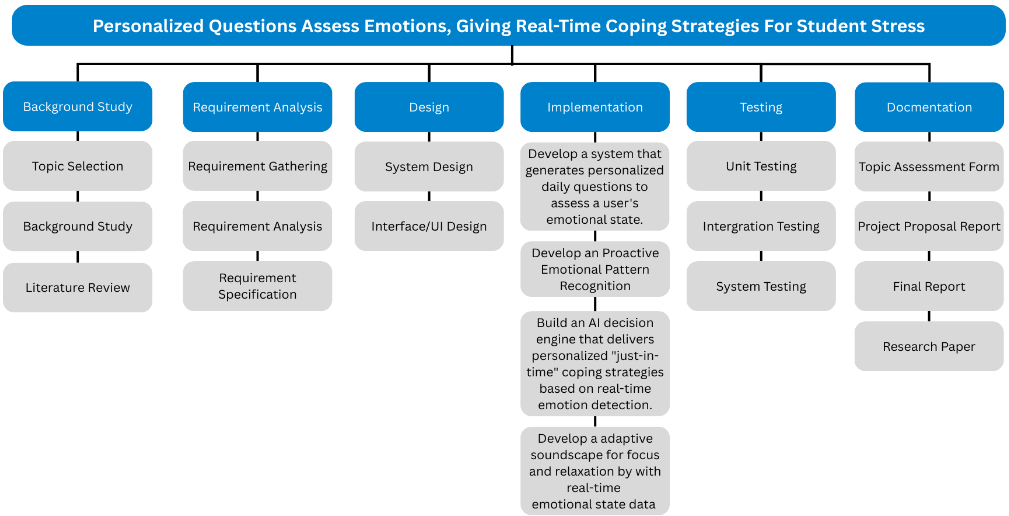

Figure 3 Work Breakdown Structure

9. PROJECT REQUIREMENTS

9.1 Functional Requirements

- The system shall present a unique, personalized question to the user each day.
- The system shall allow users to input responses via text.
- The system must analyse user input to identify and classify their emotional state.
- The system must deliver personalized coping strategies based on real-time emotional analysis.
- The system shall be capable of playing adaptive audio soundscapes.

9.2 Non-functional Requirements

- **Performance**: Real-time emotion analysis must be completed within a few seconds to provide immediate and relevant feedback.
- **Usability**: The mobile application interface must be intuitive, engaging, and easy to navigate for a young adult user base.
- **Privacy**: All user-provided data, particularly text inputs and emotional history, must be handled with strict confidentiality and stored securely.
- **Scalability**: The backend architecture must be designed to efficiently handle data processing for an increasing number of users.

9.3 User Requirements

- Help them manage academic stress proactively, not just reactively.
- Answer simple, personalized daily questions to track their emotions easily.
- Understand their feelings from their responses without requiring extra effort.
- Personalized coping strategies when stress levels are predicted to rise.

9.4 Test Cases

Table 2 Test Case1 Daily Emotional Check-In Recording

Test Case ID	TC-01	
Test Scenario	Daily Emotional Check-In Recording	
Preconditions	User is registered and has completed onboarding questionnaire. App is installed and running.	
Test Steps	**Test Input**	**Expected Results**
System triggers daily check-in notification.	-	-
User responds via text or emoji.	User input: "I feel nervous about tomorrow's exam."	Response saved under the correct user ID.
Response is stored in the database.	-	Timestamp recorded. Data linked to stress-trend visualization dashboard.

Table 3 Test Case2 Emotion Detection using NLP Model

Test Case ID	TC-02	
Test Scenario	Emotion Detection using NLP Model	
Preconditions	System has access to trained NLP model. User has submitted at least one daily response.	
Test Steps	**Test Input**	**Expected Results**
User enters a text response in the daily check-in.	User input: "I feel completely exhausted and stressed."	-
NLP model processes the input.	-	-
Classified emotion is stored with probability score.	-	Model detects High Stress with $\geq$ 85% confidence. Classification result saved in the user's profile history.

Table 4 Test Case3 Real-Time Coping Strategy Recommendation

Test Case ID	TC-03	
Test Scenario	Real-Time Coping Strategy Recommendation	
Preconditions	User has completed at least 7 days of check-ins. LSTM model predicts a stress spike within 2 days.	
Test Steps	**Test Input**	**Expected Results**
LSTM prediction engine forecasts upcoming stress escalation.	Prediction: Stress Level = High.	-
System selects coping strategy from CBT-based library..	-	-
App sends real-time notification to the user.	-	Notification sent: "You usually feel anxious before exams try this 5-minute breathing exercise." User receives at least one personalized coping recommendation.

9.5 Wireframes

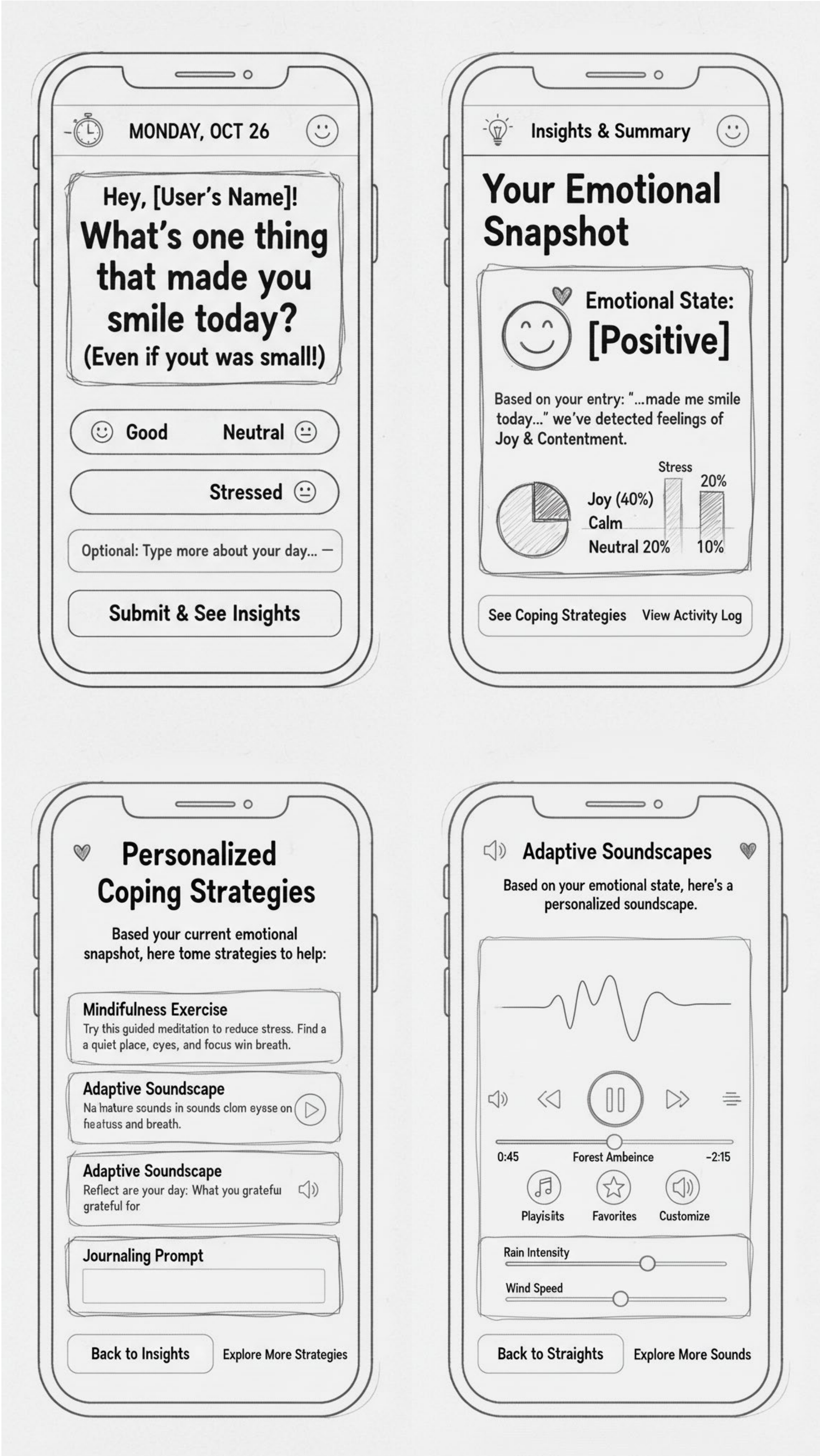

Figure 4 Wireframes

10.DESCRIPTION OF PERSONAL AND FACILITIES

Table 5 Description of personal and facilities

Registration No	Name	Task Description
IT22004390	Hettiarachchi E.I	• Design and development of the mobile application with personalized daily emotional check-ins. • Implementation of AI models (NLP + LSTM) for real-time emotion detection and stress forecasting. • Integration of CBT-based coping strategies and JITAI interventions for personalized support. • Development of adaptive soundscapes to improve student focus and relaxation. • Coordination of usability testing, model validation, and pilot evaluation with student participants. • Preparation of project documentation, research reporting, and final proposal submission.

11.BUDGET AND BUDGET JUSTIFICATION

Table 6 Estimated budget table

Item	Description	Cost (LKR)
Hardware	Mobile devices for cross-platform testing.	180,000
Cloud Services & Hosting	Firebase (Firestore, Authentication, Cloud Functions, Storage) and server costs for 12 months.	40,000
Human Resources	External Supervision (Hiring Professionals in relevant field)	25,000
Testing & Evaluation	Usability testing with students, expert validation (psychologists), and model accuracy benchmarking.	50,000
Miscellaneous	Transportation Cost	60,000
	Stationary Cost	48,000
Total		**403,000**

1. Hardware

The mobile application is being developed with React Native to ensure a consistent experience on both iOS and Android platforms. To properly test and evaluate the user-centric mobile interface, dedicated physical devices are required.

2. Cloud Services & Hosting

Ensures reliable, scalable, and secure data storage and processing. This includes Firebase services, authentication, and cloud functions for real-time operations.

3. Human Resources:

External supervision and consultation from professionals in AI, Cognitive Behavioural Therapy, and NLP are needed to validate methodologies and ensure domain-specific accuracy.

4. Testing & Evaluation

Covers usability studies with students, model accuracy testing, and validation of coping strategies with psychological experts.

5. Miscellaneous

Allocation for documentation, transportation costs, and unexpected expenses.

The total estimated budget is LKR 403,000.

12.COMMERCIALIZATION STRATEGIES

12.1 Commercialization and Revenue Opportunities

The primary goal is to create a sustainable model that ensures the application is both widely accessible to students and financially viable for long-term development and support. The following strategies leverage the unique, proactive features of the application to create distinct value propositions.

1. Freemium Model: The free version would provide essential tools for daily mental wellness management. This includes personalized daily questions to track emotional well-being, real-time emotion analysis based on text responses, Access to a limited number of fundamental coping strategies, a basic version of the adaptive soundscapes feature.
2. Premium Model: The subscription would be justified by the system's unique, proactive, and deeply personalized nature, which is a significant advancement over existing reactive apps. The proactive emotional pattern recognition and stress forecasting powered by Long Short-Term Memory (LSTM) networks. The delivery of Just-in-Time Adaptive Interventions (JITAIs) that provide the right support at the right moment.
3. Institutional Partnerships: Institutions would pay an annual licensing fee based on the size of their student population. In return, all their students would receive free, full access to the premium version of the application. It provides a proactive tool that can help prevent stress from escalating to a crisis point, complementing existing, often resource-constrained, traditional counselling services. By providing this tool, institutions demonstrate a strong commitment to student mental health, which is crucial for academic success and retention.

12.2 User Opportunities Compared to Traditional Counselling

1. Flexible Access: Students can interact with the mobile app anytime, anywhere, without being restricted by appointment schedules or physical counseling locations. This ensures continuous availability of stress support, especially during late-night study sessions and exam periods.
2. Anonymity & Comfort: The platform allows students to express their emotions and stress levels privately through daily check-ins and journaling. This anonymity reduces fear of judgment or stigma, encouraging honest responses and higher engagement than in traditional face-to-face counselling.
3. Continuous Self-Monitoring: Using real-time emotion detection and stress forecasting (via NLP and LSTM models), students can track their emotional trends and identify stress triggers proactively. Personalized coping strategies (CBT exercises, mindfulness prompts, adaptive soundscapes) and task monitoring enable measurable progress over time.

4. Peer & Institutional Support: The system can be integrated with optional university wellness services and student communities, allowing learners to receive professional guidance and peer support when required. This creates a holistic mental health ecosystem—combining digital monitoring, personalized AI-driven interventions, and human oversight where necessary.

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